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EXPERIMENT-08: Collection of Email Evidence from a Windows PC

AIM:

To capture a Windows computer's RAM using DumpIt and analyze the resulting memory image with Bulk Extractor Viewer to identify email-related artifacts.

TOOLS REQUIRED:

1. DumpIt
2. Bulk Extractor and Bulk Extractor Viewer

PROCEDURE :

Step 1: Capture RAM Using DumpIt

1. Obtain authorization to examine the Windows computer.
2. Prepare DumpIt and an external destination drive with sufficient free space.
3. Run DumpIt with the required privileges.
4. Confirm the memory acquisition prompt by entering Y if requested.
5. Wait for the acquisition to complete.
6. Record the output RAW filename, file size, acquisition time, and any reported errors.

Step 2: Locate the Memory Dump

1. Open File Explorer.
2. Navigate to the directory where DumpIt saved the output.
3. Confirm that the RAW file exists and record its size.
4. Secure the dump and calculate a SHA-256 hash where possible.

Step 3: Open Bulk Extractor Viewer

1. Obtain and install or extract a compatible version of Bulk Extractor and its viewer.
2. Launch the appropriate application.
3. Select the acquired RAW memory image as the input.
4. Choose a separate output directory for the analysis results.

Step 4: Analyze the Memory Image

1. Start the extraction process using the available options.
2. Wait for processing to finish.

3. Review the generated feature files and extraction logs.
4. Locate email.txt if the email-address feature was generated.
5. Open email_histogram.txt if available to review email-related frequency statistics.

Step 5: Record the Findings

1. Record the number of email-related artifacts found.
2. Review the extracted results and distinguish valid addresses from false positives.
3. Preserve the analysis output and document the tool version and processing status.
4. Include screenshots of the acquisition, analysis, and results in the lab record.

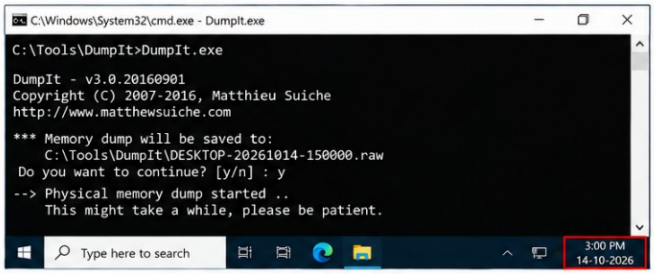


Figure 1: Running DumpIt for RAM Capture

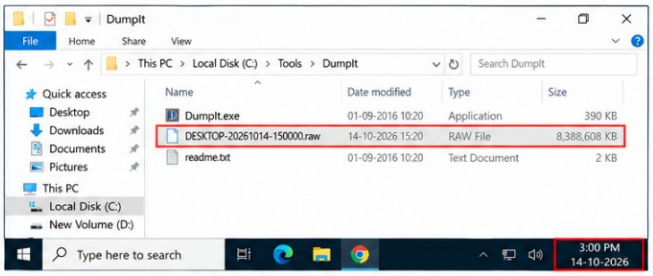


Figure 2: Generated .RAW Memory Dump File

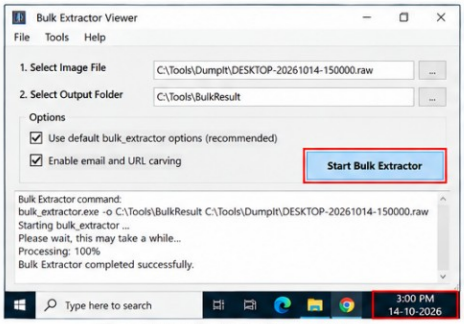


Figure 3: Running Bulk Extractor Viewer

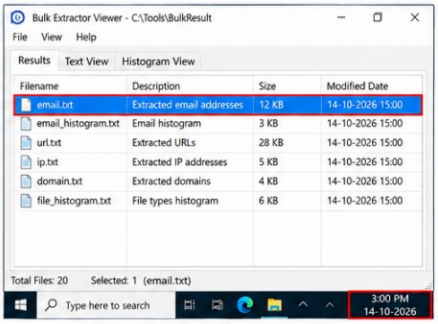


Figure 4: Analysis Results – email.txt

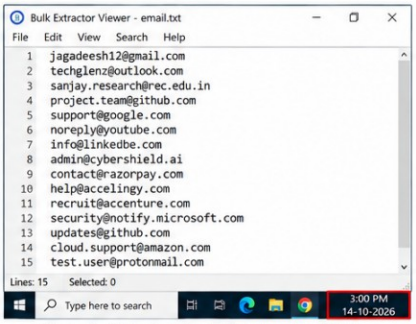


Figure 5: Extracted Email Addresses (email.txt)

RESULT:

The procedure for capturing physical memory using DumpIt and examining a memory image with Bulk Extractor Viewer was studied. Email-related artifacts were identified as a potential source of evidence.